Adapt, Generate, and Supervise: Geometry-Aware Diffusion-Guided SAM Framework for Remote Sensing Semantic Segmentation

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***Abstract*—Foundation models such as the segment anything model (SAM) have remarkable generalization capabilities in natural image segmentation. However, their application to remote sensing (RS) semantic segmentation faces significant challenges. The single-modal architecture of SAM cannot effectively utilize multimodal RS data; its vision Transformer encoder lacks sensitivity to multiscale spatial structures that are characteristic of RS imagery, and its dependence on manual prompts hinders large-scale automation. To address these challenges, we propose a geometry-aware diffusion-guided SAM framework (GeoSAM) that transforms SAM into fully automated multimodal semantic segmentation through three synergistic innovations. First, we introduce a multiscale geometric-aware adaptation module (GeoAdapter) that hierarchically integrates RGB images with normalized digital surface model (nDSM) data within the encoder. GeoAdapter incorporates a novel class-prior generator that combines geometric convolution, prototype similarity matching, and statistical modeling to produce structure-aware guidance for semantic–geometric feature fusion. Second, we present a diffusion prompt module (DPM) that pioneers the use of conditional diffusion models for automatic prompt generation in RS applications, eliminating manual interactions through semantic-feature-guided denoising diffusion implicit model (DDIM) sampling. Third, we propose a prompt-level supervision strategy (PLSS) that mitigates training–inference distribution discrepancies through constraints on semantic consistency and structural alignment, ensuring robust prompt generation across different phases. Extensive experiments demonstrate state-of-the-art performance: 90.92% mean accuracy (mAcc) and 82.71% mean intersection over union (mIoU) on Vaihingen, and 85.28% mAcc and 75.49% mIoU on Potsdam, with improvements of 16.52% mAcc and 16.29% mIoU over original SAM on Vaihingen. The source code is available at https://github.com/110-011/GeoSAM**

***Index Terms*—Diffusion model, foundational model, multimodal learning, prompt learning, remote sensing (RS), semantic segmentation.**

Received 2 October 2025; revised 17 December 2025; accepted 10 January 2026. Date of publication 13 January 2026; date of current version 29 January 2026. This work was supported in part by Zhejiang Provincial Natural Science Foundation of China under Grant ZCLZ25F0202 and in part by the National Natural Science Foundation of China under Grant 62371422 and Grant 62576317. *(Corresponding author: Caie Xu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3653952

I. Introduction

REMOTE sensing (RS) semantic segmentation is a fundamental task in Earth-observation applications that enables pixel-level understanding of complex urban planning geographical phenomena [1], environmental monitoring [2], [3], disaster assessment [4], and among others. This task assigns semantic labels to each pixel in satellite or aerial images, facilitating the automated analysis of diverse terrestrial features. However, RS images present unique computational challenges that distinguish them from natural images. These include multiscale spatial structures spanning from individual buildings to entire urban regions; rich multimodal information consisting of RGB, elevation, and hyperspectral data; complex geometric patterns with significant intraclass variations; and substantial domain shifts across different geographical locations. The complexity is further compounded by challenges in hyperspectral anomaly detection [5] and the need for innovative retrieval algorithms for multiparameter land-surface analysis [6].

Traditional RS semantic segmentation approaches have evolved significantly, from early convolutional neural networks (CNNs), such as fully convolutional networks (FCNs) [7] and end-to-end U-shaped network (UNet) [8], to more sophisticated architectures, including the pyramid scene parsing network [9]. For example, Chen et al. [10] used atrous convolution for multiscale context capture (DeepLabv3+). Vision transformers (ViTs) [11] introduced new paradigms through self-attention mechanisms, leading to hybrid CNN–transformer architectures [12], [13]. Recent advances focus on multiscale feature extraction [14], [15], multimodal fusion techniques [16], [17], [18], and specialized architectures for high-resolution imaging [19], [20], [21]. Despite these developments, existing methods face limitations regarding their generalizability across diverse RS scenarios and require extensive task-specific training data.

Foundational models have revolutionized computer vision, with the segment anything model (SAM) [22] representing a landmark achievement in general-purpose segmentation. SAM demonstrates exceptional zero-shot generalizability across diverse natural-image segmentation scenarios owing

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to its powerful ViT encoder, flexible mask decoder, and interactive prompt mechanism trained on over one billion masks. However, applying SAM directly to RS scenarios has three fundamental limitations that significantly constrain its effectiveness.

First, the SAM architecture is inherently designed for single-modal RGB input, preventing the effective exploitation of the rich multimodal information resources that are available in modern RS platforms. RS applications routinely utilize complementary modalities, including normalized digital surface models (nDSMs), digital elevation models, hyperspectral imagery, and synthetic-aperture radar data. Previous studies demonstrate that geometric information from elevation data is crucial for precise semantic understanding of RS applications. However, the single-modal design of SAM prevents it from leveraging critical complementary information sources. Second, RS imagery typically contains objects and patterns spanning multiple spatial scales, from fine-grained individual vehicles to large-scale urban districts. The ViT encoder of SAM, a powerful tool for global modeling, employs uniform patch-based processing that inadequately captures the multiscale hierarchical structures inherent in RS imagery [23]. This limitation is particularly pronounced in high-resolution satellite imagery, in which critical semantic distinctions often depend on multiscale contextual relationships. Finally, the reliance of SAM on manual prompt interactions (points, boxes, or masks) becomes impractical for large-scale RS applications that require automated processing of vast geographical regions. In contrast to interactive segmentation scenarios for natural images, RS semantic segmentation applications typically require fully automated processing workflows. This manual annotation bottleneck severely limits the scalability and practical operational RS deployment of SAM.

Recent studies attempt to address these limitations in different ways. Domain adaptation approaches [24] have shown potential for specific scenarios, and prompt engineering optimization [25] partially addresses the automation challenges. Multiscale enhancement methods [26], [27] and invariance-inspired adapters [28], [29] have been applied to overcome the spatial-structure modeling limitations. However, these approaches offer fragmented solutions that address individual limitations rather than provide a unified framework to comprehensively adapt SAM to RS requirements.

Parallel developments in diffusion models present unprecedented opportunities to address the limitations of SAM in RS applications. Recent studies demonstrate that diffusion model components built on UNet architectures effectively capture multiscale features with rich contextual and edge information, making them particularly suitable for RS image analysis. Fan et al. [30] reported promising land-use and land-cover segmentation results by combining denoising diffusion probabilistic models (DDPMs) with transformers. Although the field has witnessed increased research on diffusion model application to RS semantic segmentation tasks [31], [32], their integration with foundational models such as SAM for automated prompt generation remains largely unexplored.

To address these challenges systematically, we propose a geometry-aware diffusion-guided framework (GeoSAM), which transforms SAM from single-modal interactive segmentation into fully automated multimodal semantic segmentation through three synergistic innovations. Our approach starts with a multiscale geometric-aware adapter (GeoAdapter), which embeds hierarchical adaptation modules within the SAM encoder at strategic transformer blocks (3, 6, 9, and 12) to achieve effective multimodal feature fusion between RGB imagery and elevation data. As opposed to computationally expensive dual-stream approaches [33], our design employs efficient, direct injection while introducing a novel class-prior generator that synthesizes geometric convolution, prototype similarity matching, and statistical modeling to produce structure-aware guidance for semantic–geometric feature alignment. Building on this enhanced multimodal representation, we introduce automatic diffusion prompt generation through a diffusion prompt module (DPM), which represents the first conditional diffusion model application for automated prompt generation in RS–SAM applications. DPM leverages semantic features extracted from the enhanced SAM encoder as conditional input, generating high-quality diffusion prompts through optimized denoising diffusion implicit model (DDIM) [34] sampling, thereby eliminating manual prompt dependence while ensuring semantic accuracy. To ensure robust performance across different operational phases, we propose a prompt-level supervision strategy (PLSS), which is a dual supervision mechanism that addresses the distribution discrepancy between training (ground truth (GT)-based prompts) and inference (noise-generated prompts) phases through semantic consistency constraints and structural alignment losses, thereby ensuring robust and stable prompt generation regardless of the prompt source. This integrated approach enables GeoSAM to leverage multimodal information, automate prompt generation, and simultaneously maintain consistent performance across training and deployment scenarios.

The main contributions of this study are summarized as follows.

1) We design a novel GeoAdapter module that transforms the SAM encoder from a single-modal global modeling paradigm into a multimodal structure-aware learning framework. This is achieved by strategically inserting adaptive components and introducing a new class-prior generator that integrates RGB and elevation information through geometric convolution, prototype similarity, and statistical modeling, enabling deep semantic–structural fusion.
2) We introduce DPM, which represents the first application of conditional diffusion models for automatic prompt generation in RS–SAM applications, to achieve fully automated segmentation through semantic-feature-guided DDIM sampling.
3) We propose PLSS, a dual supervision mechanism that effectively addresses training–inference bias issues in prompt generation through semantic consistency constraints and structural alignment losses.
4) In comprehensive evaluations using the Vaihingen and Potsdam datasets, our proposed GeoSAM demonstrated state-of-the-art performance, achieving a mean accuracy (mAcc) and mean intersection over union (mIoU) of

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90.92% and 82.71%, respectively, on the Vaihingen dataset and 85.28% and 75.49%, respectively, on the Potsdam dataset.

The remainder of this article is organized as follows. Section II reviews the related work on RS semantic segmentation, SAM adaptations, and diffusion models in computer vision. Section III presents the proposed GeoSAM framework, detailing the GeoAdapter, DPM, PLSS, and feature decoder components. Section IV provides the experimental results, comparisons with state-of-the-art methods, and ablation studies. Finally, Section V concludes this article and suggests future research directions.

II. Related Work

A. RS Semantic Segmentation

The evolution of RS semantic segmentation has progressed from traditional CNN-based methods to advanced multiscale and multimodal architectures. Foundational research established the key principles. For instance, Shelhamer et al. [7] introduced the pioneering FCN architecture, enabling end-to-end learning for dense prediction tasks, while Ronneberger et al. [8] proposed UNet with skip connections, markedly improving feature preservation and boundary delineation. Zhao et al. [9] extended these approaches using atrous convolution to capture multiscale context, and Badrinarayanan et al. [35] mitigated noise in predictions caused by excessive upsampling and enhanced spatial contextual information by mapping encoder outputs to image pixel labels. Subsequent developments addressed RS-specific challenges through specialized architectures. Sun et al. [14] proposed sparse self-attention networks to efficiently leverage global and local context, whereas Li et al. [36] developed multiattention networks to capture spatial and channel dependencies. Recent advances have focused on multiscale processing [15], [27] and architectural innovations tailored to specific RS requirements [19], [20], [21]. Complementary research has explored advanced attention mechanisms for high-resolution imagery, with Wang et al. [6] proposing a hypergraph BiFormer for semantic segmentation that captures higher order spatial relationships, and Quershi et al. [38] developing semantic segmentation frameworks for aerial vehicle images. In addition, Zhong et al. [39] introduced scattering characteristics guided networks for ISAR space target component segmentation, demonstrating the versatility of attention-based approaches across diverse RS applications.

Although single-modal approaches have achieved notable progress, they are inherently limited by sensitivity to lighting variations, weather conditions, and incomplete scene representation. This limitation has shifted research toward exploiting complementary information from multisensor platforms [40]. With the growing availability of multimodal RS data, significant advances have been made in cross-modal feature-fusion techniques. However, traditional fusion methods—whether early fusion through input-level concatenation or late fusion combining separate network outputs—often fail to capture the complex cross-modal relationships essential for accurate semantic understanding. To address this, researchers have drawn inspiration from advanced fusion strategies in the broader computer-vision community. For example, Hu et al. [16] proposed the attention complementary network (ACNet), which selectively integrates features from RGB and depth branches, demonstrating effective cross-modal attention mechanisms. Chen et al. [17] introduced bidirectional cross-modal feature propagation, enabling sophisticated feature interactions between modalities. Building on these multimodal learning principles, transformer-based approaches [18], [41] have been successfully adapted to RS contexts, demonstrating superior capability in modeling long-range dependencies across sensor modalities [42]. Recent advances have further demonstrated the effectiveness of cross-modal feature integration, with Wang et al. [43] proposing cross-modal segmentation networks for winter-wheat mapping in complex terrain using multitemporal images and digital elevation model data, and Zhang et al. [44] developing the spectral–spatial dual-branch fusion transformers for hyperspectral image classification, both highlighting the importance of effective multimodal fusion strategies in RS applications.

Despite these advances, existing RS semantic segmentation methods have fundamentally limited generalizability and depend on manual annotation. These constraints underscore the urgent need for foundational model-based approaches that can leverage pretrained knowledge while adapting to the unique characteristics of RS data.

B. Segment Anything Model

SAM demonstrates strong zero-shot generalization through large-scale pretraining and prompt-driven segmentation [22]. However, directly applying SAM to RS semantic segmentation remains challenging owing to three intrinsic limitations: its single-modal RGB design, limited sensitivity to multiscale geospatial structures, and strong dependence on manually specified prompts. These issues have motivated a series of SAM-based adaptations for RS imagery [23].

Several methods address these limitations from different perspectives. To improve adaptation efficiency, Chen et al. [29] proposed injecting lightweight adaptation modules into the SAM encoder to enable parameter-efficient fine-tuning (SAM-Adapter), while related adapter-based designs further explore invariance-aware parameter modulation strategies [28]. Although effective in reducing training costs, these approaches operate solely on RGB features and lack explicit modeling of elevation-derived or geometric priors, which are crucial for distinguishing structurally similar classes in RS scenes. From a structural perspective, Zhou et al. [26] introduced a multiscale enhancement strategy to strengthen SAM's perception of hierarchical spatial patterns (MeSAM), and Li et al. [28] further incorporated multiscale decoding mechanisms to improve spatial consistency (MDSAM) [27]. While these methods enhance scale-aware feature aggregation, they remain focused on structural representation and do not incorporate category-aware geometric semantics or multimodal information. In addition, several studies explore multimodal extensions or weakly supervised adaptations of SAM, such as dual-stream designs for multimodal inputs [33], foundation models for optical–radar fusion [45], and uncertainty-guided or

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boundary-constrained SAM variants [46], [47], [48]. These approaches either introduce substantial computational overhead or still rely on heuristic or manually initialized prompts.

While the above methods typically address adaptation efficiency or structural perception in isolation, GeoSAM provides a unified solution. Unlike the parameter-efficient adaptation strategy proposed by Chen et al. [29], GeoSAM explicitly injects geometry-aware, category-sensitive priors into the SAM encoder, enabling semantic–geometric coupling beyond feature-level modulation. Compared with the multiscale enhancement strategy introduced by Zhou et al. [26], GeoSAM goes beyond scale-aware representation by leveraging elevation-derived information to guide class-specific learning. Furthermore, GeoSAM eliminates manual prompt dependence by introducing a diffusion-based automatic prompt-generation mechanism, making it more suitable for fully automated and large-scale RS semantic segmentation.

C. Diffusion Models in Computer Vision and RS

Diffusion models have transformed computer vision by introducing a powerful generative-modeling paradigm that learns complex data distributions through iterative denoising. Originally implemented via DDPMs [49], these models reverse the forward diffusion that gradually corrupts data with Gaussian noise, enabling high-quality sample generation through learned denoising functions. Their theoretical foundation is based on stochastic differential equations and score-based generative modeling, and has been further advanced by deterministic sampling approaches, such as DDIM [34], which improve computational efficiency, and stable diffusion methods [50], which enable scalable high-resolution synthesis.

Building on these advances, applying diffusion models to semantic segmentation represents a shift from traditional discriminative methods to generative frameworks that model underlying distributions of segmentation masks. Wu et al. [51] pioneered this approach using UNet-based architectures to synthesize segmentation maps, demonstrating that the multiscale feature extraction capabilities of diffusion models—arising from their encoder–decoder structures with skip connections—are particularly well suited for dense prediction tasks. This opened new avenues for conditional generation in segmentation, where diffusion models leverage conditional inputs to produce pixel-level annotations [52], [53].

These capabilities in semantic segmentation have naturally extended to RS applications, in which the strengths of diffusion models in terms of distributional modeling align well with the unique challenges of RS imagery, such as domain variations across geographical regions and imaging conditions [30]. While discriminative approaches learn decision boundaries, diffusion models learn the underlying data distribution, enabling more robust generation under the domain variations that are commonly encountered in RS imagery across different geographical regions, seasons, and imaging conditions. This distributional modeling capability has been exploited in recent years. For example, Xie et al. [54] leveraged diffusion models as sophisticated data augmenters for instance segmentation, whereas Tang et al. [55] explored the potential of diffusion models in object detection through diffusion-driven synthetic data generation techniques. More recently, Li et al. [56] investigated federated-learning paradigms that integrate diffusion models for multimodal RS applications [31], [32], showcasing the increasing recognition of the versatility of these models in handling complex RS scenarios.

Despite these advances, the integration of diffusion models with foundation models such as SAM for automated prompt generation in RS applications remains largely unexplored, presenting a significant opportunity for innovation. This synergy is compelling: the conditional generation capabilities of diffusion models can mitigate the dependence of SAM on prompts, while their multiscale feature extraction via UNet architectures aligns with RS requirements for hierarchical spatial understanding. However, their conditional framework naturally supports the integration of diverse RS data modalities, including elevation models and optical imagery. By addressing critical gaps in automated prompt generation, training–inference consistency, and multimodal integration, the proposed GeoSAM framework leverages a novel diffusion-based prompt module to unlock the full potential of diffusion models in RS semantic segmentation, enabling more robust and automated workflows.

III. Proposed Method

A. Overview

The proposed GeoSAM framework addresses the fundamental limitations of SAM in RS applications through a unified end-to-end architecture that transforms single-modal interactive segmentation into fully automated multimodal semantic segmentation. The core innovation lies in synergistically integrating the critical GeoAdapter, DPM, and PLSS components, as illustrated in Fig. 1. GeoSAM uses given RGB (I_{rgb}) and nDSM (I_{ndsm}) features as inputs. First, the framework generates a class prior using a class-prior generator, which leverages geometric, prototype-based, and statistical modeling to produce a structure-aware guidance prior, P_{prior}. Subsequently, the GeoAdapter-enhanced ViT encoder processes the RGB features, while hierarchically injecting geometric information from the nDSM and class priors at specific transformer blocks {3, 6, 9, 12}, producing multiscale feature representations, F_i (i = 3, 6, 9, 12). Simultaneously, DPM extracts semantic conditioning vectors from the final encoder features and employs a conditional diffusion process, which automatically generates a diffusion prompt, P_{diff}. During training, this process uses GT masks with added noise to learn data denoising, whereas during inference, DDIM sampling is employed for efficient prompt generation. The prompt encoder encodes generated prompts to produce dense prompt embeddings, which, combined with positional encodings, guide the two-way transformer decoder to generate intermediate features F_{decoder}. Finally, the feature decoder executes hierarchical feature fusion by combining multiscale encoder and decoder features through progressive upsampling and skip connections, ultimately producing class-specific predictions using specialized decoder branches.

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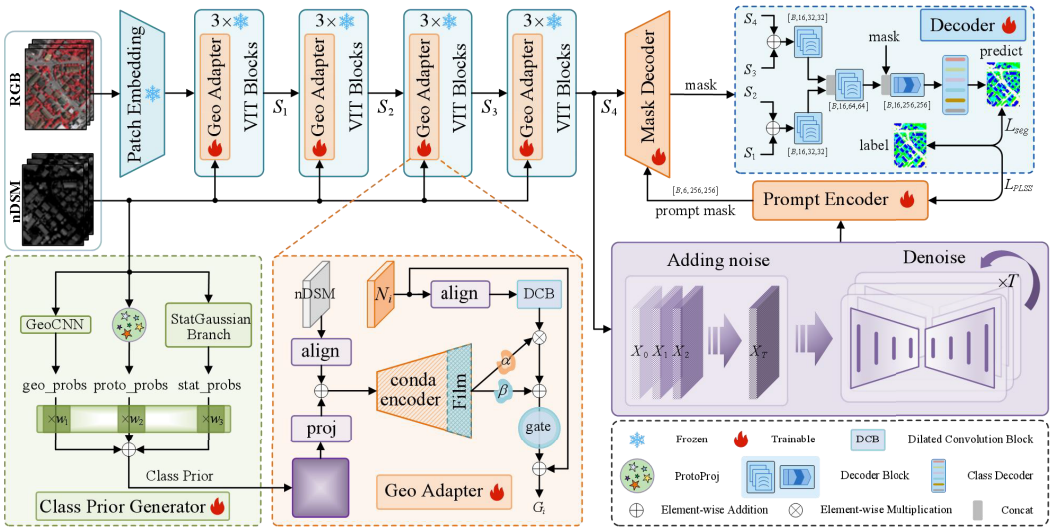


Fig. 1. Overview of the GeoSAM architecture.

B. GeoAdapter

The GeoAdapter module is designed to enhance the performance of SAM's ViT encoder with multimodal feature fusion and multiscale geometric perception, enabling the network to effectively integrate complementary RGB and nDSM information. Its architecture comprises two key components: a class-prior generator and a feature-modulation unit strategically embedded in specific transformer blocks (3, 6, 9, and 12), which progressively incorporate geometric information at different semantic scales. These specific blocks {3, 6, 9, 12} are strategically selected to correspond to feature resolutions of {128, 64, 32, 16} pixels, providing hierarchical coverage from fine-grained geometric details to global semantic patterns while maintaining computational efficiency through uniform spacing. This design enables the model to simultaneously capture local geometric structures in shallow layers and high-level semantic concepts in deeper layers, facilitating effective multiscale geometric–semantic fusion throughout the encoding process.

At its core, the class-prior generator of the GeoAdapter establishes a structured relationship between elevation patterns and semantic categories through three complementary mechanisms. First, the phenomenon "semantic categories exhibit unique geometric characteristics in elevation data" exists; for instance, buildings have typical height profiles, and trees display natural variations in elevation, while low-growing vegetation types maintain relatively homogenous surfaces. This geometric–semantic correspondence can be modeled as a learned mapping from the surface morphology to category probabilities. Through a geometric branch that employs convolutional layers specifically designed for elevation data, we extract structural features, such as building height profiles and vegetation canopy morphology

$$P_{\text{geo}} = \text{Softmax}\left(\text{Conv}_{\text{geo}}\left(I_{\text{ndsm}}\right)\right) \tag{1}$$

where Softmax$(\cdot)$ denotes the softmax activation function, and $\text{Conv}_{\text{geo}}(\cdot)$ consists of a cascaded structure comprising a 3×3 convolution, rectified linear unit activation, and a 1×1 convolution.

Furthermore, based on the metric learning theory, we propose that each semantic category can be represented by a learnable prototype in a shared geometric–semantic embedding space. Despite intraclass variations, the similarity between the observed features and category prototypes provides reliable classification guidance. Therefore, in the prototype similarity branch, we maintain a learnable prototype ω_c for each semantic category c. By computing the similarity scores between the projected features and these prototypes, we offer geometrically informed classification guidance, maintaining robustness even with intraclass variations

$$F_{\text{proj}} = \text{Conv}_{1\times 1}\left(I_{\text{ndsm}}\right) \tag{2}$$

$$\text{Sim} = \frac{F_{\text{proj}} \cdot \omega_c}{\left\|F_{\text{proj}}\right\|_2 \cdot \left\|\omega_c\right\|_2} \tag{3}$$

$$P_{\text{proto}} = \text{Softmax}\left(\text{Sim}\right) \tag{4}$$

where Sim represents the similarity tensor between the projected features and class prototypes.

We simultaneously employ a statistical normalization approach that adapts to category-specific distributions to address the inherent variability of semantic categories across different geographical regions and handle intraclass geometric variations. For each class, we maintain running statistics μ_c and σ_c^2, which are updated through momentum-based estimation during training, ensuring robustness when geometric features undergo domain shifts and providing resilience against geometric variability

$$P_{\text{stat}}\left(x, c\right) = \frac{1}{\sqrt{2\pi\sigma_c^2}} \exp\left(-\frac{(x-\mu_c)^2}{2\sigma_c^2}\right) \tag{5}$$

where μ_c and σ_c^2 represent the running mean and variance for class c, respectively,

$$\mu_c^{t+1} = (1-\delta)\,\mu_c^t + \delta\mu_c^{\text{batch}} \sigma_c^{t+1} = (1-\delta)\,\sigma_c^t + \delta\sigma_c^{\text{batch}} \tag{6}$$

with momentum parameter $\delta = 0.01$.

Then, the outputs of the three branches are adaptively weighted and combined into a structure-aware class prior,

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which serves as semantic–geometric guidance for subsequent feature modulation

$$P_{\text{prior}} = w_1 \cdot P_{\text{geo}} + w_2 \cdot P_{\text{proto}} + w_3 \cdot P_{\text{stat}} \tag{7}$$

where w_i $(i = 1, 2, 3)$ is the learnable parameter adapted based on the scene characteristics.

These class priors are then injected into the SAM encoder using a hierarchical geometric injection strategy, which is based on the multiscale characteristics of RS features, where shallow layers capture fine geometric details and deeper layers encode semantic concepts. We achieve progressive geometric–semantic fusion with minimal parameter overhead by strategically deploying adapters in blocks 3, 6, 9, and 12. Each adapter corresponds to different spatial resolutions (128, 64, 32, and 16 pixels) and distinct semantic granularities, enabling the model to simultaneously capture local geometric details and global semantic patterns. Furthermore, the core conditional modulation mechanism within the adapter employs feature-wise linear modulation (FiLM) to achieve semantically aware feature adjustment

$$F_{\text{cond}} = \text{Encoder}_{\text{cond}}(I_{\text{ndsm}}) + \text{Proj}_{\text{class}}(P_{\text{prior}}) \tag{8}$$

$$\gamma, \beta = \text{split}(\text{Conv}_{\text{film}}(F_{\text{cond}})) \tag{9}$$

$$F_{\text{adapted}} = \gamma \cdot \text{DCB}(\text{Down}(F_i)) + \beta \tag{10}$$

where DCB$(\cdot)$ applies dilated convolutions with rates {2, 4} to capture the multiscale context, and the FiLM parameters γ and β provide adaptive feature scaling and shifting. Encoder$_{\text{cond}}(\cdot)$, Proj$_{\text{class}}(\cdot)$, Conv$_{\text{film}}(\cdot)$, and Down$(\cdot)$ each comprise a 1×1 convolutional block, and $i = 3, 6, 9, 12$.

Finally, the adapted features are integrated through a gated residual connection

$$F_{\text{output}} = F_i + \rho \cdot \text{UP}(F_{\text{adapted}}) \tag{11}$$

where ρ is a learnable gate parameter that controls the adaptation strength, and UP$(\cdot)$ is the upsampling operation.

C. Diffusion Prompt Module

The explicit dependence of SAM on prompts severely constrains its application in large-scale RS scenarios, while traditional prompt-generation methods (such as saliency-based algorithms or simple thresholding) fail to capture complex semantic relations in RS imagery. DPM pioneers the introduction of conditional diffusion models into RS scene analysis (SAM) applications, achieving fully automated prompt generation via semantic-feature-guided denoising processes. This module not only eliminates dependence on manual prompts but also ensures high-quality category-aware segmentation guidance.

The adoption of conditional diffusion models over lightweight CNN segmentation heads for prompt generation is grounded in their fundamentally different learning paradigms. While CNNs learn discriminative decision boundaries through one-shot prediction, diffusion models learn the underlying distribution $P(X|\text{condition})$, enabling the generation of diverse high-quality prompts that capture multimodal uncertainty in ambiguous RS regions. The iterative denoising process allows progressive semantic conditioning at multiple scales, while CNNs may overlook subtle semantic cues critical for accurate prompts. However, this distributional approach facilitates training–inference consistency through PLSS, as diffusion models seamlessly handle both label-based and noise-based inputs, whereas CNN predictors trained solely on label supervision suffer significant distribution-gap degradation.

First, to fully exploit the rich semantic features of the SAM encoder, the global semantic context is extracted from the final layer features of the SAM encoder through global average pooling (GAP)

$$f_{\text{semantic}} = \text{GAP}(F_{\text{final}}) \tag{12}$$

where F_{final} represents the final layer features of the enhanced SAM.

Subsequently, the diffusion timestep t is encoded using sinusoidal positional encoding and fused with the semantic features to generate the conditional semantic features required by DPM

$$\text{TimeEmbed}(t) = [\sin(t \cdot w), \cos(t \cdot w)] \tag{13}$$

$$f_{\text{cond}} = f_{\text{semantic}} + \text{MLP}(\text{TimeEmbed}(t)) \tag{14}$$

$$w = \exp\left(-\log(10\,000) \cdot \frac{\text{arrange}(D/2)}{D/2 - 1}\right) \tag{15}$$

where $D = 128$ is the embedding dimension, and MLP$(\cdot)$ denotes a multilayer perceptron. This design ensures that the diffusion process is aware of both the semantic content and the denoising process.

Finally, the DPM core employs a lightweight conditional diffusion network based on a SegFormer backbone (MiT-B0) [57], which integrates the UNet architecture and FiLM conditioning mechanisms to extract multimodal semantic features from the SAM encoder, generating diffusion prompts that regulate the operational state of DPM. Each UNet layer incorporates FiLM blocks for conditional modulation

$$\varepsilon_\theta(X_t, t, f_{\text{cond}}) = \text{UNet}(X_t, \text{TimeEmbed}(t), f_{\text{cond}}) \tag{16}$$

$$\text{FiLM}(f, f_{\text{cond}}) = \gamma(f_{\text{cond}}) \cdot \text{Conv}_{3\times3}(f) + \beta(f_{\text{cond}}) \tag{17}$$

where X_t is the noisy diffusion prompt at timestep t, f_{cond} denotes the conditional semantic features, ε_θ is the predicted noise, f represents the UNet intermediate layer features, and conv$_{3\times3}$ represents a 3×3 convolutional block.

During training, DPM learns to denoise corrupted GT masks. The forward diffusion process is a Markov chain in which, given the original prompt map (GT mask) X_0, at each timestep $t \in \{1, \ldots, T\}$, Gaussian noise is added to generate X_t

$$X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon \tag{18}$$

where $\varepsilon \in N(0, I), \varsigma_t \in (0, 1)$ is the noise step (linear scheduling) and $\alpha_t = 1 - \varsigma_t$, and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative retention rate.

To enhance the training objective, in addition to the standard DDPM loss function, we introduce variational information distillation (VID) [58] to improve the training stability

$$L_{\text{diff}} = E_{t, X_0, \varepsilon}\left[\|\varepsilon - \varepsilon_\theta(X_t, t, c)\|_2^2\right] + L_{\text{VID}} \tag{19}$$

where L_{VID} is the VID loss

$$L_{\text{VID}} = E\left[\frac{1}{2}\left(\log(\phi^2) + \frac{(\varepsilon - \varepsilon_\theta)^2}{\phi^2}\right)\right] \tag{20}$$

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with a learnable variance parameter $\phi^2 = \log(1+\exp(\tau))+\xi$. τ is initialized to achieve $\varphi^2 = 5$, and ξ represents the minimum value to prevent the denominator from becoming zero.

During the inference phase, we no longer use the label X_0, but instead start from noise and efficiently generate a diffusion prompt through DDIM deterministic sampling

$$X_{t-\nabla_t} = \sqrt{\bar{\alpha}_{t-\nabla_t}}\left(\frac{X_t - 1 - \sqrt{\bar{\alpha}_t}\varepsilon_\theta(X_t, f_{\text{cond}}, t)}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1-\bar{\alpha}_{t-\nabla_t}-\psi_t^2}\varepsilon_\theta(X_t, t, c) + \psi_t\varepsilon_t \quad (21)$$

where $\psi_t = \eta\cdot((1-\bar{\alpha}_{t-\nabla_t})/(1-\bar{\alpha}_t))^{1/2}\cdot(1-\bar{\alpha}_t/\bar{\alpha}_{t-\nabla_t})^{1/2}$. The physical meaning lies in its deterministic reverse diffusion process. Unlike the standard DDPM that introduces stochasticity at each step, DDIM establishes a direct correspondence between the noise prediction ε_θ and the denoised sample $X_{t-\nabla t}$. The term $(\bar{\alpha}_{t-\nabla_t})^{1/2}$ scales the predicted clean prompt toward the data manifold, while $(1-\bar{\alpha}_{t-\nabla_t})^{1/2}\varepsilon_\theta(X_t, f_{\text{cond}}, t)$ represents the residual noise component aligned with the conditional semantic features. By setting $\eta = 0$, this formulation enables deterministic sampling that maintains semantic consistency across multiple inference runs while significantly reducing computational cost from 1000 to 50 steps, making automated prompt generation practical for large-scale RS applications.

D. Feature Decoder

The feature decoder serves as the final synthesis component, which is responsible for combining multiscale encoder features with prompt-conditioned decoder features to generate accurate semantic predictions. First, to address the semantic gap between features at different scales, we adopt a hierarchical fusion strategy that groups and processes the encoder features. The core insight is that different encoder stages capture complementary information—the early stages preserve the spatial details, while the deeper stages encode the semantic concepts. Subsequently, we apply a hierarchical combination approach for the progressive feature fusion of the processed grouped features, integrating prompt-conditioned decoder features through skip connections

$$F_{\text{early}} = \text{DecoderBlock}(S_1 + S_2) \quad (22)$$

$$F_{\text{deep}} = \text{DecoderBlock}(S_3 + S_4) \quad (23)$$

$$F_{\text{fused}} = \text{DecoderBlock}\left(\text{Concat}\left(F_{\text{early}}, F_{\text{deep}}\right)\right) \quad (24)$$

$$F_{\text{final}} = \text{Conv}_{3\times3}(\text{Concat}(F_{\text{fused}}, F_{\text{mask}})) \quad (25)$$

where DecoderBlock$(\cdot)$ consists of a cascaded structure comprising a transpose convolutional block, BatchNorm, and rectified linear unit activation; Concat$(\cdot)$ denotes the concatenation operation; S_i $(i = 1, 2, 3, 4)$ represents the stratified feature; F_{mask} represents the SAM mask decoder feature.

Finally, we adopt a category-specific decoding strategy to address the multiclass problem in RS semantic segmentation. As opposed to using a single classification head, we design category-specific decoders in which each semantic category has a dedicated branch. The final segmentation map P is obtained by concatenating the category-specific prediction results from each branch along the channel dimension

$$\begin{cases} M_1 = \text{Decoder}_1(F_{\text{final}}) \\ M_2 = \text{Decoder}_2(F_{\text{final}}) \\ \vdots \\ M_3 = \text{Decoder}_3(F_{\text{final}}) \end{cases} \quad (26)$$

$$P = \text{Concat}([M_1, M_2, \ldots, M_c]) \quad (27)$$

where each Decoder$(\cdot)$ consists of an independent convolutional block. This design enables each decoder branch to specialize in specific semantic patterns, improving the overall segmentation quality.

E. Prompt-Level Supervision Strategy

PLSS aims to address a critical challenge in this framework: the training–inference distribution discrepancy caused by different prompt sources. This discrepancy arises because DPM generates prompts by receiving GT masks during the training phase, whereas it generates prompts directly from noise during the inference phase. This inconsistency can lead to performance degradation and unstable segmentation results. Therefore, in the PLSS strategy, we employ two complementary supervision signals to ensure that the training system produces stable and consistent results regardless of the prompt sources.

First, to ensure semantic consistency between prompts generated from different sources and the GT labels, we introduce the cross-entropy and dice losses as semantic consistency losses

$$L_{\text{sem}} = \text{CrossEntropy}\left(P_{\text{gen}}, Y\right) + \text{Dice}\left(P_{\text{gen}}, Y\right) \quad (28)$$

$$\text{CrossEntropy}(P, Y) = -\frac{1}{HW}\sum_{i,j,c} Y_{i,j,c}\log\left(P_{i,j,c}\right) \quad (29)$$

$$\text{Dice}(P, Y) = 1 - \frac{2\sum_{i,j,c} P_{i,j,c}Y_{i,j,c} + \xi}{\sum_{i,j,c} P_{i,j,c} + \sum_{i,j,c} Y_{i,j,c} + \xi} \quad (30)$$

where H and W represent the height and width of P_{gen} and Y, respectively. P_{gen} represents the generated diffusion prompt, and Y denotes the GT labels.

In addition to semantic consistency constraints, we compute the $L2$ norm between P_{gen} and Y as the structural alignment loss to ensure that the generated prompts preserve important geometric and structural information

$$L_{\text{struct}} = \left\|P_{\text{gen}} - Y\right\|_2^2 + \left\|\nabla P_{\text{gen}} - \nabla Y\right\|_2^2. \quad (31)$$

The gradient term is used to maintain boundary sharpness, which is crucial for precise segmentation.

F. Total Loss Function

The total loss used in the proposed framework consists of three components, namely the main task segmentation loss L_{seg}, diffusion loss L_{diff}, and PLSS loss $L_{\text{PLSS}} = L_{\text{sem}} + L_{\text{struct}}$. For L_{seg}, we introduce the cross-entropy and dice losses to supervise the discrepancy between the predicted results

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P output by GeoSAM and the GT labels Y. The specific formulation is as follows:

$$L_{\text{seg}} = \text{CrossEntropy}\left(P_{\text{gen}}, Y\right) + \text{Dice}\left(P_{\text{gen}}, Y\right) \tag{32}$$

$$L_{\text{total}} = L_{\text{seg}} + L_{\text{diff}} + L_{\text{PLSS}}. \tag{33}$$

IV. Experiments and Results

A. Datasets and Evaluation Metrics

1) Datasets: We conducted extensive experiments on two large-scale datasets, namely the Vaihingen and Potsdam datasets, to validate the performance of the proposed GeoSAM model. Both datasets followed the same classification scheme, encompassing six major categories: low vegetation, impervious surfaces, buildings, vehicles, trees, and clutter/background. The Vaihingen dataset contained 33 high-resolution three-channel RGB images of 2500 × 2000 pixels with a corresponding nDSM at a spatial resolution of 9 cm. To ensure experimental fairness, we adopted the recommended parameter settings for the dataset: 17 images (numbered 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38) were selected as the test set, five images (numbered 11, 15, 28, 30, and 34) were selected as the validation set, and the remaining 11 images were used as the training set. The Potsdam dataset comprised 38 three-channel RGB images of 6000 × 6000 resolution with a corresponding nDSM at a spatial resolution of 5 cm. Following the officially recommended parameters for dataset partitioning, 31 images were selected for the training and testing sets, with the remaining seven images (numbered 2_11, 2_12, 4_10, 5_11, 6_7, 7_8, and 7_10) serving as the validation set.

2) Evaluation Metrics: Throughout the experimental process, we employed multiple quantitative metrics: overall accuracy (Acc), mAcc, intersection over union (IoU), and mIoU, to evaluate the performance of GeoSAM accurately.

B. Implementation Details

All experiments were conducted using the PyTorch framework on a personal computer equipped with an NVIDIA TITAN XP GPU (12 GB of memory). During the experiments, images from the Potsdam and Vaihingen datasets were cropped to 256 × 256 pixels and augmented with vertical and horizontal flips as inputs, with a batch size of 12 for 200 iterations. We employed the AdamW optimizer with a weight decay coefficient of $1e^{-4}$ for training. The SAM encoder parameters were frozen, while the remaining components of the original SAM were set with a learning rate of $5e^{-5}$. Our proposed modules were configured with a learning rate of $5e^{-4}$.

C. Results and Analysis

We compared our proposed GeoSAM with 17 existing models: traditional CNN architectures (FCN-8 S, DeepLabv3+, SegNet, ACNet, and SA-Gate), advanced transformer architectures (CMXNet and CATNet), recent SAM-based improvements (SAM, MeSAM, MDSAM, and SAM-Adapter), and other state-of-the-art methods (SDANet, MSGCNet, AFENet, SFAF-MA, CPSSNet, and UNetMamba). The prompt encoding module of the original SAM was removed and fully fine-tuned for RS semantic segmentation. To ensure a fair comparison, the segmentation results for all models were generated using either the original data provided by the authors or their pretrained models. All evaluation metrics were computed using unified evaluation tools.

1) Quantitative Analysis: Tables I and II present the quantitative results for GeoSAM and the comparative methods on the Vaihingen and Potsdam datasets, respectively. Overall, the proposed GeoSAM achieved the best performance on both datasets: 90.92% mAcc and 82.71% mIoU on the Vaihingen dataset, and 85.28% mAcc and 75.49% mIoU on the Potsdam dataset. Compared with the original SAM, GeoSAM demonstrated significant improvements, with mAcc and mIoU increasing by 16.52% and 16.29%, respectively, on the Vaihingen dataset, and by 15.11% and 14.81%, respectively, on the Potsdam dataset. Furthermore, an individual category quantitative analysis revealed that GeoSAM performed exceptionally well in challenging scenarios, particularly in vehicle detection (improving the mAcc and mIoU by 7.97% and 3.57%, respectively, over the best competing model on the Vaihingen dataset) and tree extraction (achieving 80.53% and 73.35% mIoU on the Vaihingen and Potsdam datasets, respectively). Despite integrating multiple components, GeoSAM maintained reasonable computational efficiency, with an inference time of approximately 0.082 s per 256 × 256 image and 0.005 GB memory consumption.

2) Qualitative Analysis: Fig. 2 presents the segmentation maps predicted using GeoSAM and other methods for representative samples from both datasets. In complex RS scenarios, GeoSAM precisely segmented objects at different scales while completely recognizing their geometric structures. Compared with other approaches, our proposed GeoSAM demonstrated superior advantages in terms of detailed representations and boundary accuracy, particularly in building extraction and vegetation-impervious surface delineation areas, where rough boundaries or fragmentation issues commonly occur. Furthermore, GeoSAM exhibited exceptional performance in handling multiscale targets: it accurately detected and segmented small vehicles that were frequently missed by the baseline methods while maintaining consistent classification of large urban structures and avoiding over-segmentation. In addition, by integrating nDSM to acquire elevation information, our method effectively preserved the geometric relationships between different semantic categories, ensured spatial constraint authenticity, and achieved more coherent segmentation results than traditional methods. Despite differences in spatial resolution (9 versus 5 cm) and scene characteristics between Vaihingen and Potsdam, GeoSAM consistently delivered performance improvements across both datasets, fully demonstrating its robust generalizability.

D. Ablation Studies

We conducted comprehensive ablation studies on both datasets to validate the effectiveness of each proposed component rigorously and understand its individual contributions. Our analysis systematically examined the impact of the three core components: GeoAdapter, DPM, and PLSS, with a particular focus on their logical dependencies and synergistic effects.

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TABLE I

QUANTITATIVE COMPARISON ON THE VAIHINGEN DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND GREEN, RESPECTIVELY

Models	Imp.surf		Building		Low veg		Tree		Car		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S[7]	89.66	79.71	93.22	86.80	75.83	64.33	89.22	75.58	45.12	40.16	78.61	69.32
SegNet[35]	89.88	80.93	90.88	86.54	78.66	64.07	88.96	75.96	43.93	43.16	78.46	70.13
DeepLabv3+[9]	90.06	81.11	87.04	82.70	76.65	64.44	88.60	76.64	42.51	43.10	76.97	69.49
ACNet[16]	91.95	85.34	95.45	91.82	78.64	66.87	91.20	78.55	83.12	76.81	88.07	79.88
SA-Gate[17]	90.99	85.70	93.85	91.72	84.95	68.68	89.06	79.15	84.27	78.07	88.62	80.67
SDANet[14]	91.62	81.60	91.72	88.01	79.24	67.38	90.55	78.58	70.69	68.73	84.76	76.86
CMXNet[18]	91.78	85.39	95.15	91.28	77.00	66.29	92.59	78.85	83.23	78.58	87.95	80.08
SFAF-MA[41]	90.94	85.37	95.67	92.21	80.99	67.55	90.78	78.95	74.74	71.19	86.62	79.05
SAM[22]	90.83	81.14	94.85	90.71	77.31	64.64	89.29	76.30	19.75	19.30	74.40	66.42
SAM-Adapter[29]	91.43	85.74	95.65	92.25	79.85	67.78	91.04	78.26	76.03	71.64	86.80	79.13
CATNet[13]	90.53	83.98	95.27	91.08	78.27	65.62	90.57	77.72	70.28	61.85	84.98	76.05
MSGCNet[15]	91.20	84.92	94.93	90.94	77.73	67.04	92.34	78.19	85.06	78.69	88.25	79.95
MDSAM[27]	92.80	85.51	95.79	92.61	79.02	69.69	93.33	80.61	84.26	79.60	89.04	81.60
MeSAM[26]	91.58	85.73	95.53	92.29	80.44	67.24	90.17	78.89	85.11	80.28	88.57	80.89
UNetMamba[21]	91.98	85.13	95.71	92.40	77.79	67.16	92.13	79.04	84.43	79.20	88.41	80.60
AFENet[37]	91.88	84.52	93.96	91.02	80.29	67.66	91.03	78.59	82.94	78.92	88.02	80.14
CPSSNet[20]	93.28	83.28	95.31	90.71	71.28	64.44	94.02	79.04	83.05	77.26	87.39	78.95
GeoSAM	92.27	86.58	95.73	92.57	81.29	70.00	92.26	80.53	93.08	83.85	90.92	82.71

TABLE II

QUANTITATIVE COMPARISON ON THE POTSDAM DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND GREEN, RESPECTIVELY

Models	Imp.surf		Building		Low veg		Tree		Car		Cluter		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S[7]	89.47	79.77	90.69	83.60	85.13	71.12	82.86	71.23	91.02	81.53	49.05	36.49	78.61	69.32
SegNet[35]	90.18	80.46	90.21	84.18	85.88	71.63	82.49	70.68	93.15	89.72	51.76	37.21	82.28	72.31
DeepLabv3+[9]	91.57	82.49	91.78	87.59	87.36	73.63	85.45	73.32	93.89	90.04	53.80	43.54	83.97	75.10
ACNet[16]	91.32	82.74	93.83	90.06	86.16	73.53	86.03	72.87	93.79	90.43	54.51	41.65	84.27	75.21
SA-Gate[17]	85.84	80.64	93.65	88.51	86.46	72.71	85.70	72.89	92.18	89.39	62.70	40.59	84.42	74.12
SDANet[14]	90.03	81.68	90.87	85.75	88.17	73.18	82.02	71.32	93.94	89.82	56.72	40.03	83.63	73.63
CMXNet[18]	90.54	81.86	93.80	89.28	87.17	73.11	83.52	71.45	92.23	87.99	51.61	40.58	83.14	74.04
SFAF-MA[41]	90.63	82.06	93.31	89.67	85.76	72.35	83.51	70.95	92.55	88.34	60.03	42.40	84.30	74.30
SAM[22]	87.39	79.53	90.37	86.59	82.28	70.25	83.91	70.67	33.64	35.76	43.45	21.25	70.17	60.68
SAM-Adapter[29]	88.30	79.91	91.82	88.70	83.14	70.21	84.96	68.83	91.16	84.35	51.70	36.25	81.85	71.38
CATNet[13]	88.12	79.80	93.32	89.52	85.91	70.66	78.78	68.85	91.56	84.99	55.58	40.41	82.21	71.87
MSGCNet[15]	86.96	78.25	91.49	82.45	82.71	70.86	82.96	67.07	95.88	76.30	56.39	39.81	82.73	69.46
MDSAM[27]	91.55	82.89	93.55	89.79	87.88	73.40	83.40	72.38	93.55	88.11	51.36	40.49	83.55	74.51
MeSAM[26]	88.84	81.88	91.61	88.26	87.54	72.84	83.50	72.35	93.64	89.70	53.80	39.96	83.15	74.16
UNetMamba[21]	90.67	83.00	94.07	89.53	86.34	73.45	86.31	73.21	93.69	89.61	54.11	42.21	84.20	75.17
AFENet[37]	91.52	82.18	90.93	86.52	87.26	73.63	83.87	72.97	94.53	90.89	56.84	42.08	84.16	74.71
CPSSNet[20]	88.87	81.23	91.82	85.61	89.02	73.18	81.38	71.95	93.23	89.56	54.76	38.48	83.18	73.34
GeoSAM	89.45	81.76	92.84	89.57	84.92	72.18	88.37	73.35	94.98	91.87	61.09	44.24	85.28	75.49

TABLE III

ABLATION EXPERIMENT ON CORE COMPONENTS

Components	VAIHINGEN		POTSDAM	
	mAcc	mIoU	mAcc	mIoU
Baseline	74.40	66.42	70.17	60.68
SAM with GeoAdapter	88.58	80.53	83.33	73.76
SAM with DPM	88.19	80.57	83.26	73.58
SAM with GeoAdapter and DPM	89.73	81.32	84.16	73.89
SAM with DPM and PLSS	89.57	81.22	83.98	73.92
GeoSAM (complete)	90.92	82.71	85.28	75.49

1) Core Component Ablation Analysis: This study validated the effectiveness of each core component through systematic ablation experiments. The quantitative results are presented in Table III, and visual comparisons are shown in Figs. 3–6. The baseline represents the original SAM architecture adapted for RS semantic segmentation, with the prompt encoding module removed and full fine-tuning applied.

a) Ablation on GeoAdapter: SAM with GeoAdapter alone demonstrated significant performance improvements on the Vaihingen dataset, with improvements of 14.18% and 14.11% in mAcc and mIoU, respectively, compared with the baseline, achieving 88.58% mAcc and 80.53% mIoU. As shown in Fig. 3, SAM with the GeoAdapter-alone configuration exhibited notable improvements in boundary delineation and reduced fragmentation, particularly in complex geometric structures. This demonstrates the critical importance of multimodal geometric perception and multiscale feature adaptation in RS scenarios, especially when the elevation data provide crucial disambiguation between semantically similar but geometrically distinct objects.

b) Ablation on DPM: SAM with the DPM-alone component also demonstrated individual performance comparable to GeoAdapter, with improvements of 13.79% and 14.15% in mAcc and mIoU, respectively, on the Vaihingen dataset, achieving 88.19% mAcc and 80.57% mIoU. Fig. 4 clearly illustrates that SAM with DPM alone significantly enhanced the segmentation consistency and reduced false positive detections compared with the baseline, validating that

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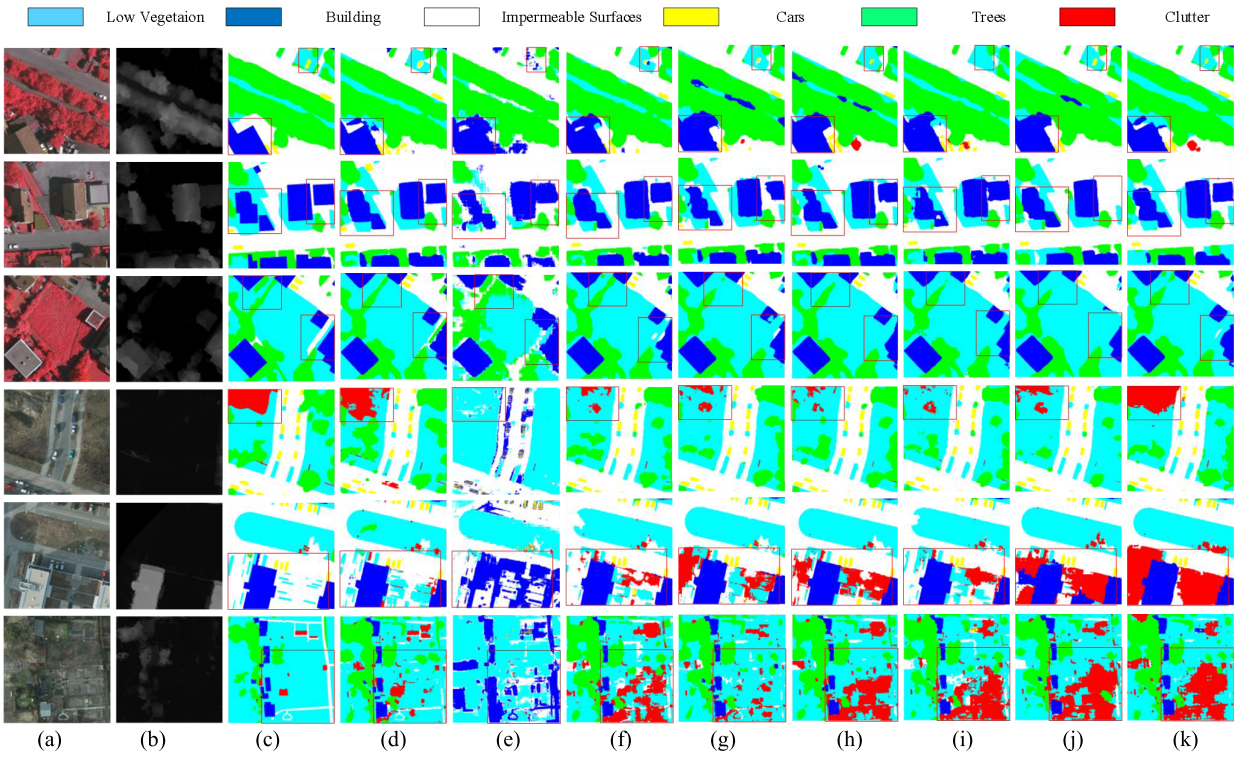


Fig. 2. Semantic segmentation results: (a) RGB, (b) nDSM, (c) GT, (d) GeoSAM (ours), (e) SAM, (f) SAM-Adapter, (g) MeSAM, (h) MDSAM, (i) SFAF-MA, (j) SDANet, and (k) SA-Gate. For ease of observation, red boxes are used to mark the prediction's salient areas.

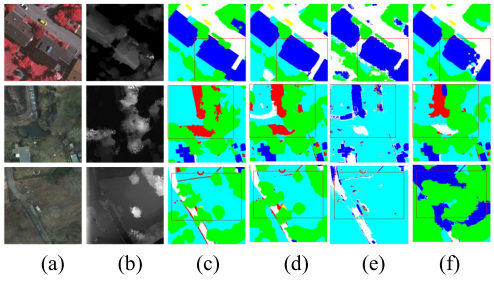


Fig. 3. Comparison of the results of GeoAdapter: (a) RGB, (b) nDSM, (c) GT, (d) GeoSAM, (e) baseline (SAM), and (f) SAM with GeoAdapter.

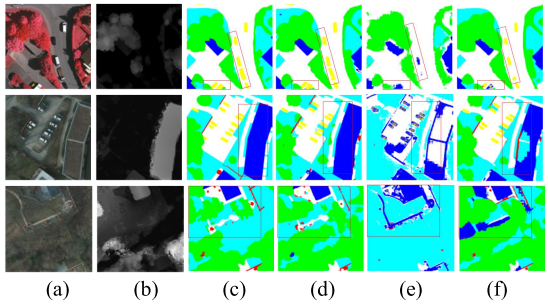


Fig. 4. Comparison of the results of DPM: (a) RGB, (b) nDSM, (c) GT, (d) GeoSAM, (e) baseline (SAM), and (f) SAM with DPM.

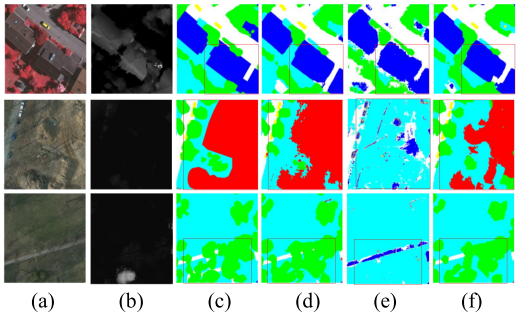


Fig. 5. Comparison of combined results of GeoAdapter and DPM: (a) RGB, (b) nDSM, (c) GT, (d) GeoSAM, (e) baseline (SAM), and (f) SAM with GeoAdapter and DPM.

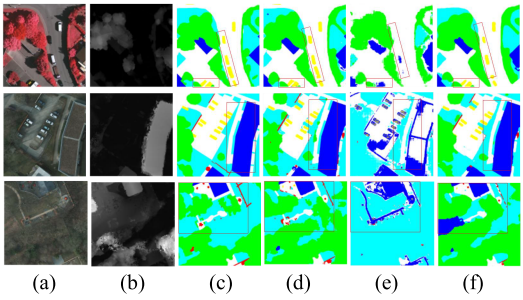


Fig. 6. Comparison of combined results of DPM and PLSS: (a) RGB, (b) nDSM, (c) GT, (d) GeoSAM, (e) baseline (SAM), and (f) GeoSAM with DPM and PLSS.

diffusion-based automatic prompt generation can effectively guide the segmentation process without manual intervention.

c) Ablation on the GeoAdapter and DPM combination: The SAM, GeoAdapter, and DPM combination yielded superior performance compared with individual components, achieving 89.73% mAcc and 81.32% mIoU on Vaihingen, and 84.16% mAcc and 73.89% mIoU on Potsdam. As shown in Fig. 5, the SAM, GeoAdapter, and DPM combination produced more coherent segmentation results with enhanced spatial consistency, with the structure-aware feature modulation enhancing the semantic conditioning for

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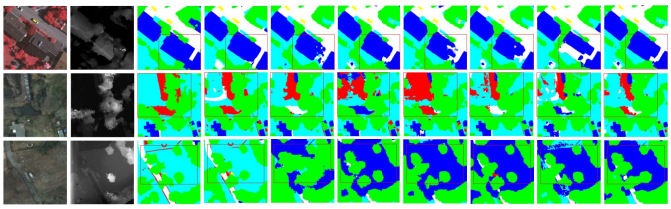


Fig. 7. Comparison of the results of GeoAdapter internal structure: (a) RGB, (b) nDSM, (c) GT, (d) GeoSAM, (e) GeoAdapter (complete), (f) GeoAdapter with geometric branch, (g) GeoAdapter with prototype similarity, (h) GeoAdapter with statistical modeling, (i) GeoAdapter with geometric branch and prototype similarity, and (j) GeoAdapter with geometric branch and statistical modeling.

TABLE IV
INDIVIDUAL COMPONENT VALIDATION OF GEOADAPTER

Method	VAIHINGEN		POTSDAM	
	mAcc	mIoU	mAcc	mIoU
GeoAdapter with geometric branch	87.21	79.15	82.39	72.38
GeoAdapter with prototype similarity	87.53	79.34	82.45	72.57
GeoAdapter with statistical modeling	87.42	79.21	82.41	72.65
GeoAdapter with geometric branch and prototype similarity	87.69	79.54	82.57	72.76
GeoAdapter with geometric branch and statistical modeling	87.70	79.43	82.62	72.89
GeoAdapter with prototype similarity and statistical modeling	87.98	79.82	82.87	73.00
GeoAdapter(complete)	88.58	80.53	83.33	73.76

diffusion-generated prompts, while the high-quality automated prompts facilitated more accurate geometric–semantic alignment, demonstrating positive synergistic effects between multimodal geometric adaptation and automatic prompt generation.

d) Ablation on the DPM and PLSS combination: The PLSS component is specifically designed to address the training–inference distribution discrepancy caused by different prompt sources in the diffusion process. Therefore, it was only evaluated in combination with DPM. On the Vaihingen dataset, the SAM, DPM, and PLSS combination achieved substantial improvements of 1.38% and 0.65% in mAcc and mIoU, respectively, compared with SAM with DPM alone, reaching 89.57% mAcc and 81.22% mIoU. A comparison between Figs. 4 and 6 clearly demonstrate the visual impact of PLSS: SAM with DPM and PLSS produced more stable and consistent segmentation results with reduced noise and improved boundary precision. This targeted improvement validates our theoretical analysis that PLSS effectively mitigates bias introduced by using GT masks during training versus noise-generated prompts during inference.

2) GeoAdapter Internal Structure Ablation: To gain deeper insights into the GeoAdapter design rationale, we conducted detailed ablation studies on its internal components, focusing on the class-prior generator and feature modulation mechanisms. Table IV and Fig. 7 present the ablation results for the three core components of the class-prior generator. This analysis validated our hypothesis that different geometric–semantic relationships require complementary modeling approaches.

Each branch contributed substantially to the overall segmentation performance. On the Vaihingen dataset, GeoAdapter with a geometric branch alone improved mIoU from 66.42% (baseline) to 79.15%, confirming the importance of capturing the spatial structure via elevation-based patterns that distinguish buildings from vegetation. GeoAdapter with a prototype similarity branch yielded a similar performance of 79.34% mIoU, validating the role of learnable class-specific geometric representations in semantic disambiguation. GeoAdapter with a statistical modeling branch achieved 79.21% mIoU, indicating that modeling intraclass variance enhances robustness to geometric variations across different geographical regions.

When combined, these branches exhibited clear complementarity. GeoAdapter with a geometric branch and prototype similarity reached 79.54% mIoU, GeoAdapter with a geometric branch and statistical modeling achieved 79.43% mIoU, and GeoAdapter with prototype similarity and statistical modeling achieved the highest dual-configuration performance of 79.82% mIoU. This suggests that learnable prototypes benefit significantly from adaptive statistical normalization. Notably, GeoAdapter (complete) with all three branches achieved the best performance: 80.53% mIoU and 88.58% mAcc on Vaihingen, and 73.76% mIoU and 83.33% mAcc on Potsdam. These results support our hypothesis that different geometric–semantic relationships require diverse modeling approaches, and their integration provides stronger structural guidance during feature modulation. Fig. 7 visually demonstrates these progressive improvements, showing enhanced boundary precision and reduced classification errors in geometrically complex regions in which multiple semantic categories intersect.

TABLE V
ABLATION EXPERIMENT ON ADAPTATION MECHANISMS

Method	VAIHINGEN		POTSDAM	
	mAcc	mIoU	mAcc	mIoU
Full fine-tuning	74.40	66.42	70.17	60.68
Film	86.79	78.61	82.25	72.54
LoRA	86.53	78.52	81.89	71.96
SAM-Adapter	86.80	79.13	81.85	71.38
GeoAdapter	88.58	80.03	83.33	73.76

To validate our GeoAdapter design, Table V provides a comparison with established adaptation mechanisms, including the LoRA [59], FiLM [60], and standard adapter [29] approaches.

All adaptation methods significantly outperformed full fine-tuning, which achieved only 66.42% mIoU on Vaihingen. Among the baseline methods, FiLM conditioning reached 78.61% mIoU, demonstrating effective FiLM for conditional adaptation with an improvement of 12.19% over full fine-tuning. LoRA adaptation achieved 78.52% mIoU, validating parameter-efficient adaptation through low-rank matrix decomposition, with an improvement of 12.10%. The standard adapter mechanism achieved 79.13% mIoU, showing the benefits of dedicated adaptation modules within transformer architectures, with an improvement of 12.71%. However, our GeoAdapter substantially surpassed all comparison methods, achieving 80.03% mIoU and 88.58% mAcc on Vaihingen, which represents a 0.9% mIoU improvement over the best competitor (standard adapter). In Potsdam, GeoAdapter achieved 73.76% mIoU and 83.33% mAcc, maintaining consistent superiority across different datasets.

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TABLE VI
ABLATION EXPERIMENT ON DDIM SAMPLING STEPS

DDIM Steps	Vaihingen		Potsdam		Inference Time (s)	Memory (GB)
	mAcc	mIoU	mAcc	mIoU		
10	89.23	81.45	83.71	74.12	0.032	0.003
25	90.15	82.18	84.89	74.87	0.058	0.004
50	90.92	82.71	85.28	75.49	0.082	0.005
100	90.94	82.73	85.31	75.52	0.147	0.007
200	90.96	82.75	85.33	75.54	0.265	0.009

TABLE VII
ABLATION EXPERIMENT ON CONDITIONAL INFORMATION

Conditioning Strategy	Vaihingen		Potsdam	
	mAcc	mIoU	mAcc	mIoU
Time encoding only	87.42	79.83	82.15	69.92
Semantic features only	88.76	80.79	83.84	72.58
Complete conditioning	90.92	82.71	85.28	75.49

3) Verification of Key DPM Design Choices: We conducted systematic ablation studies on the DDIM sampling steps and conditional information design to validate the critical design decisions in our DPM.

a) Ablation on DDIM sampling steps: Table VI presents the performance–efficiency trade-off across different DDIM sampling steps, revealing a clear pattern of diminishing returns as the sampling steps increase. With only ten steps, DPM achieved a reasonable baseline performance of 81.45% mIoU on Vaihingen with minimal computational overhead, requiring a 0.032-s inference time and 0.003 GB of memory, thereby establishing the lower bound for practical prompt generation. The performance improved substantially with 25 steps, reaching 82.18% mIoU with a gain of 0.73%, and furthermore, with 50 steps, 82.71% mIoU was achieved, representing an overall improvement of 1.26% over the ten-step baseline.

However, increasing the sampling steps further yielded rapidly diminishing returns. At 100 steps, the performance reached 82.73% mIoU with only a marginal 0.02% improvement over 50 steps, while the inference time almost doubled to 0.147 s. Similarly, 200 steps provided only 82.75% mIoU with a total gain of 1.30% over ten steps, but required a 0.265-s inference time—more than eight times that of the minimal configuration. These results clearly justify our choice of 50 steps as the optimal configuration, achieving 96.90% of the maximum performance while maintaining practical computational efficiency. The 50-step setting captures the essential denoising dynamics without the computational overhead of excessive sampling iterations.

b) Ablation on conditional information design: As shown in Table VII, the importance of our complete conditional information design was validated through systematic ablation on the conditioning components. Using only timestep encoding achieved a moderate performance of 79.83% mIoU on Vaihingen, demonstrating that temporal awareness alone provides basic denoising guidance but lacks semantic context for class-aware prompt generation. Semantic features alone substantially improved the performance to 80.79% mIoU, confirming that global semantic context from the SAM encoder provides crucial class-specific information for generating meaningful prompts. This validates our hypothesis that semantic conditioning is more critical than temporal information in automated prompt generation in RS scenarios.

However, the complete conditioning strategy that fuses both timestep encoding and semantic features achieved optimal performance, reaching 82.71% mIoU and 90.92% mAcc on Vaihingen. This represents improvements of 2.16% in mAcc and 1.92% in mIoU over semantic-only conditioning, demonstrating clear complementarity between temporal and semantic information. The results confirm that our complete conditional feature design is essential in high-quality prompt generation, in which semantic context guides what to generate while temporal encoding informs how to generate it progressively through the denoising process.

TABLE VIII
ABLATION EXPERIMENT ON EACH PLSS SUPERVISION STRATEGY

Conditioning Strategy	Vaihingen		Potsdam	
	mAcc	mIoU	mAcc	mIoU
Without PLSS	89.73	81.32	83.98	73.92
PLSS with $L_{semantic}$ only	89.96	81.44	84.46	74.70
PLSS with L_{struct} only	89.89	81.68	84.27	74.49
Complete PLSS	90.92	82.71	85.28	75.49

4) PLSS Supervision Strategy Effectiveness: To evaluate the effectiveness of PLSS in mitigating the training–inference distribution discrepancy, we conducted a systematic ablation on its constituent loss components. The results are presented in Table VIII. Without PLSS, the baseline achieved 89.73% mAcc and 81.32% mIoU on Vaihingen, and 83.98% mAcc and 73.92% mIoU on Potsdam.

In closing, we first examine the contribution of the semantic consistency loss (L_{semantic}), which enforces classwise alignment between the generated prompt masks and GT labels using the cross-entropy and dice losses. PLSS with L_{semantic} yielded measurable gains of 0.23% mAcc and 0.12% mIoU on Vaihingen, reaching 89.96% mAcc and 81.44% mIoU, and improvements of 0.48% mAcc and 0.78% mIoU on Potsdam, reaching 84.46% mAcc and 74.70% mIoU. These improvements confirm that promoting semantic fidelity during training helps DPM to produce more accurate and discriminative prompts. The structural alignment loss (L_{struct}), which applies $L2$ regularization and gradient-based boundary constraints, also provided notable enhancements. PLSS with L_{struct} improved the performance by 0.16% mAcc and 0.36% mIoU on Vaihingen, reaching 89.89% mAcc and 81.68% mIoU, and by 0.29% mAcc and 0.57% mIoU on Potsdam, reaching 84.27% mAcc and 74.49% mIoU. These results validate that preserving geometric structures and boundary sharpness is essential in maintaining spatial coherence in generated prompt masks.

When both loss terms were jointly applied, the complete PLSS supervision achieved the best performance, reaching 90.92% mAcc and 82.71% mIoU on Vaihingen, and 85.28% mAcc and 75.49% mIoU on Potsdam. These results represent a total gain of 1.90% mAcc and 1.39% mIoU on Vaihingen, and 1.30% mAcc and 1.57% mIoU on Potsdam compared with the model without PLSS, respectively. The consistent improvements across both datasets highlight the complementary nature of semantic and structural constraints: the former

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ensures precise class discrimination, while the latter maintains geometric integrity and boundary alignment.

The systematic ablation on PLSS components demonstrated that both the semantic consistency and structural alignment losses contribute meaningfully to addressing the training–inference distribution gap. Their complementary effects validate our design rationale that effective prompt learning supervision requires both semantic accuracy and geometric fidelity to bridge the discrepancy between GT training masks and noise-generated inference prompts.

E. Discussion on Limitations and Future Directions

Despite the strong performance of GeoSAM, several limitations warrant discussion. The current framework is specifically designed for RGB–nDSM fusion and requires architectural modifications to incorporate additional modalities, such as synthetic-aperture radar or hyperspectral data. The iterative denoising process in DPM, although optimized to 50 DDIM steps, still incurs higher computational overhead compared with direct prediction methods, potentially limiting real-time applications in time-critical scenarios. In addition, performance on extremely large-scale satellite scenes (e.g., exceeding $10\,000 \times 10\,000$ pixels) may be constrained by memory limitations and the fixed patch size of 256×256 pixels, potentially requiring sliding-window strategies that could introduce boundary artifacts.

These limitations suggest several promising future directions. Phase congruency-based feature extraction [61] could improve edge preservation and multimodal alignment, particularly for low-overlap aerial imagery scenarios. Prior information constraints [62] could enhance the semantic consistency of generated prompts while reducing dependence on large-scale annotated datasets. Advanced attention mechanisms, such as hypergraph-based approaches [63], could capture higher order relationships among multimodal features, improving segmentation of complex urban spatial patterns. Multimodal remote perception learning strategies [64] could enable more robust feature representations across diverse sensor modalities and imaging conditions, addressing domain shift challenges in global-scale applications. Furthermore, self-supervised pretraining on large-scale unlabeled RS imagery could improve domain generalization and reduce fine-tuning requirements for new geographical regions, advancing toward universal RS semantic segmentation systems.

V. Conclusion

This study advances the adaptation of foundation segmentation models to RS by highlighting the importance of explicitly integrating domain-specific geometric information and reducing reliance on manual intervention. Rather than focusing solely on architectural modifications or parameter-efficient tuning, the proposed framework demonstrates a principled direction for bridging generic vision models with the unique structural and semantic characteristics of geospatial imagery.

From a broader perspective, this study suggests a scalable paradigm for applying foundation models to large-area RS analysis, with potential extensions beyond semantic segmentation to other geospatial understanding tasks. Future research may explore more expressive geometric representations and richer multimodal integration strategies to further enhance robustness and generalization, contributing to the development of more autonomous and reliable Earth observation systems.

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